

Advanced Observability Lab on GCP — Complete Guide

Overview:

This lab demonstrates a production-grade Observability Pipeline using:

- Node.js application instrumented with OpenTelemetry SDK
- OTLP HTTP exporter
- OpenTelemetry Collector
- Google Cloud Trace
- Two GCP VMs inside same VPC

Architecture:

User → app-vm (Node.js + OTEL SDK) → OTLP → monitoring-vm (OTEL Collector) → Cloud Trace

What We Achieved:

- Two GCP VMs created
- Collector installed & receiving traces
- Node.js app auto-instrumented
- Traces visible in Cloud Trace
- End-to-end distributed tracing pipeline

Production Use Cases:

- Microservices tracing
- RCA and debugging
- APM
- SRE Observability
- Performance insights

Step-by-Step Implementation:

1. Create two GCP VMs (monitoring-vm & app-vm)
2. Install OpenTelemetry Collector

3. Configure OTLP receivers & Cloud Trace exporter
4. Install Node.js & dependencies
5. Create server.js app
6. Install OTEL auto-instrumentations
7. Configure tracing.js
8. Run Node application
9. Generate traffic
10. Validate traces in Cloud Trace

Files Included:

- server.js
- tracing.js
- config.yaml
- Architecture diagram
- README.md